A Medication Adherence Promotion System to Reduce Late Kidney Allograft Rejection: A Quality Improvement Study

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Rationale & Objective: Adolescent and young adult kidney transplant recipients have a high risk of rejection related to suboptimal adherence. Multicomponent interventions improve adherence in controlled trials, but clinical implementation is lacking. We describe an initiative to reduce allograft rejection using evidence-based adherence promotion strategies.

Study Design: Interrupted time series.

Setting & Participants: Kidney transplant recipients cared for at Cincinnati Children's Hospital ≥ 1 year after transplant and taking ≥1 immunosuppressive medication(s) from 2014 through 2017.

Quality Improvement Activities: The following interventions, collectively called MAPS (Medication Adherence Promotion System), were implemented over 14 months: (1) adherence promotion training for clinical staff, 2) electronic health record–supported adherence risk screening, (3) systematic assessment of medication adherence barriers, (4) designation of specific staff to address adherence barriers, (5) shared decision-making with the patients to overcome adherence barriers, (6) follow-up evaluation to assess progress, and (7) optional electronic medication monitoring.

Outcomes: Primary Outcome: Late acute rejection. Process measures were conducted to assess barriers, identify barriers, and perform interventions. The secondary outcomes/balancing measures were de novo donor-specific antibodies (DSA), biopsy rate, and rejections per biopsy.

Analytical Approach: Time series analysis using statistical process control evaluated patient-days between acute rejections as well as monthly rejections per 100 patient-months before and after implementation. To control for known rejection risk factors including changes in treatment and case mix, multivariable analyses were performed.

Results: The monthly rejection rate fell from 1.61 rejections per 100 patient-months in the 26 months before implementation to 0.88 rejections per 100 patient-months in the 22 months after implementation. In the multivariable analysis, MAPS was associated with a 50% reduction in rejection incidence (incidence rate ratio, 0.50 [95% CI, 0.27-0.91]; $P = 0.02$). DSA and time since transplant (per each additional year) were also associated with rejection incidence (incidence rate ratio, 2.27 [$P = 0.02$] and 0.87 [$P = 0.02$], respectively).

Limitations: Single-center study, and potential confounding by unmeasured variables.

Conclusions: Clinical implementation of evidence-based adherence-promotion strategies was associated with a 50% reduction in acute rejection incidence over 2 years.

Complete author and article information provided before references.

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Preventing allosensitization is the primary objective of care after solid-organ transplantation.[1] Yet despite potent immunosuppression, nonadherence commonly undermines treatment and results in rejection. For kidney transplant (KT) recipients, nonadherence and resulting antibody[2] or T-cell-mediated[3] rejection (AMR and TCMR, respectively) are the strongest predictors of allograft failure. Adolescents and young adults are disproportionately affected[4] by suboptimal adherence because of normal developmental changes that impact adherence behaviors (eg, brain development, independence, risk taking, and peer influence).[5] Consequently, they experience allograft failure at 3 times the rate of others,[6,7] which often requires them to return to dialysis, with a 25-year decrease in their life expectancy[8] and worsened quality of life[9] for much higher cost.[10]

Adherence barriers are amenable to behavioral interventions,[11] and provider-delivered, systems-based, multicomponent interventions have demonstrated large and durable effects.[12,13] Three randomized trials in solid-organ transplantation have demonstrated the efficacy of such adherence interventions,[11,14,15] yet clinical implementation has been lacking.[16] Further, although improved adherence should logically decrease allograft rejection, clinical trials have been underpowered to evaluate this end point.[11,14-16]

In 2015, our interdisciplinary team began a quality improvement (QI) initiative to decrease late acute rejection in our KT population by implementing evidence-based adherence strategies as standard care. As in the published clinical trials,[11,14,15] these strategies directly address adherence barriers, which correlate highly with adherence behaviors.[17,18] We applied structured QI methods to

PLAIN-LANGUAGE SUMMARY

Children, adolescents, and young adults face many barriers to taking medication as prescribed. After an organ transplant, failure to address these barriers can lead to transplant rejection and organ failure. In our kidney transplant clinic, we implemented an approach to help patients identify barriers to taking medication and to partner with clinical team members to overcome them. After full implementation of the new system in our clinic over 15 months, acute rejection in our patients decreased by nearly half. These results need to be replicated by studies in other clinics and settings, but this approach may help clinicians know how to improve health outcomes by partnering with patients who have chronic illness to overcome the barriers to taking medication.

develop, refine, and implement interventions as part of clinic workflows, and we tracked rejection rates and process measures over time using statistical process control (SPC) and run charts.

Methods

We followed the SQUIRE 2.0 guidelines[19] for reporting. The Cincinnati Children's Hospital (CCHMC) Institutional Review Board approved this study with a waiver of consent (study ID: 2017-6122).

Setting

The CCHMC KT program performs 20 to 30 transplants annually and cares for ~120 posttransplant patients. After the first year, patients are seen in the clinic quarterly, with monthly laboratory monitoring. Most patients receive dual immunosuppression with tacrolimus and either mycophenolate mofetil or azathioprine. Patients with higher immunologic risk also receive low-dose prednisone. Donor-specific antibodies (DSAs) are assessed using Luminex assays before and after transplant at 3, 6, and 12 months and then annually. Mean fluorescence intensity cutoffs of 1,500 are considered positive. All biopsies after 1 year are "for cause." Our team consists of 9 pediatric nephrologists, 6 nephrology fellows, 3 transplant nurse coordinators, and an advanced practice nurse, pharmacist, social worker, clinical psychologist, and dietitian. Before the QI intervention, we implemented systems for recording each rejection within the electronic health record (EHR), care management by transplant nurse coordinators, and EHR-assisted previsit planning where the team reviews patients' records before their clinic visits.

The analysis period spanned January 1, 2014, through December 31, 2017. All patients were included who were >1 year posttransplant and were taking at least one immunosuppressant medication at any time during the study period. The majority of patients were observed during both the pre- and post-QI intervention periods, and a single patient could contribute multiple rejections during either phase or both. New transplant patients or patients who transferred into the cohort were included on the date of their first transplant anniversary or their transfer, whichever was later. Patients who transferred care to another center, experienced allograft failure, developed malignancy, or died were excluded from analysis on the event date.

Interventions

In October 2014, after approval by clinical leadership for this QI activity, our interdisciplinary team (including a software engineer and former patient) convened to identify essential system components to systematically identify and address patient barriers to taking immunosuppression (Fig 1; Table 1). This bundle of interventions, collectively called the Medication Adherence Promotion System (MAPS), includes the following: (1) adherence promotion training for clinical team members, (2) EHR-supported adherence risk screening,[20] (3) systematic assessment of barriers to taking medication (Fig 2),[21] (4) designation of specific team members responsible to partner with patients to address barriers (Fig 2), (5) patient-centered shared decision-making to develop action plans (see Table 2), (6) planned follow-up evaluation to assess progress and adapt action plans, and (7) optional electronic adherence monitoring.

Implementation and Study of Interventions

Each intervention component was iteratively tested, refined, and implemented sequentially using plan-do-study-act cycles, as previously described elsewhere.[21] We started testing each intervention component with 1-2 patients across 1-2 providers and then, after refinement, spread the intervention to additional patients and providers until it was fully implemented as standard care for all visits. We started with paper barriers assessments in January 2015.[21] By the end of February 2016, all MAPS components had been integrated into clinical workflow care (Fig 1).

Measures

The primary outcome measure was biopsy-proven late acute rejection, acute TCMR grade 1a or higher or AMR according to the Banff 2013 criteria.[22] There was a single instance of fulminant kidney failure assumed to be rejection without biopsy confirmation; this occurred in a patient who had not come for care in many months. Since 2011, every rejection episode was documented in discrete fields within the EHR. Data and biopsy reports were extracted from the EHR and were manually validated by the first and second authors. Biopsies with rejection within 60 days of a prior rejection in the same patient were considered persistent rejection whereas rejections > 60 days apart were considered separate events. Because this

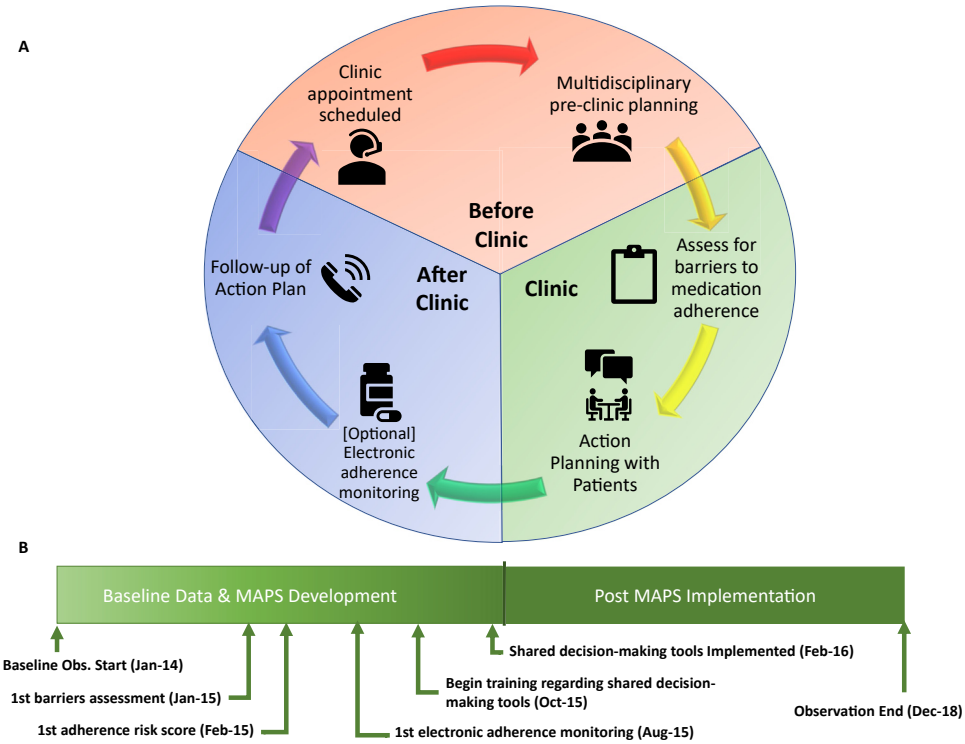


Figure 1. The medication adherence promotion system components and implementation timeline. (A) MAPS is designed to integrate with the cycle of patient care and address actions before during and after the clinic visit in a continuous cycle. Before a scheduled clinic appointment, the multidisciplinary team reviews relevant data including adherence risk. During check-in for their clinic appointment, patients and/or caregivers are given a tablet computer to complete an assessment for barriers to taking medication. Answers are transmitted directly to the clinical encounter in the EHR, so the clinician is aware of any identified barriers and can develop an action plan with the patient as detailed in Table 1 to overcome the identified barrier. Higher risk patients are offered optional electronic monitoring for outside the clinic, and any action plans that were implemented are followed up on by telephone. This cycle continues indefinitely. (B) Timeline for implementation of each component of the system during the study period. Abbreviations: EHR, electronic health record; MAPS, Medical Adherence Promotion System; Obs., observation.

was a clinical QI initiative, adherence was not directly measured using electronic monitors in most patients. Rather, we focused on the clinical outcome of rejection—the primary objective of adherence strategies in transplantation.

Process measures assessed the percentage of eligible patients who completed a barriers assessment tool during their clinic visit[21] and the number of adherence barriers and interventions documented in the population. To assess whether changes in rejection rates were related to unintentional decreases in "for cause" biopsies, we assessed the incidence rate of kidney biopsies and biopsies positive for rejection before and after MAPS implementation. Secondary outcomes included the incident rate of de novo DSA development and the type (TCMR, AMR, mixed) and severity of rejection before and after MAPS implementation.

Both time-invariant and time-varying covariates were analyzed as part of the patient-level analysis. The time-invariant covariates included biological sex, race/ethnicity, human leukocyte antigen (HLA) mismatch, induction immunosuppression (missing for some patients transplanted at other centers), donor type, and primary or secondary transplant. The time-varying covariates included history of prior rejection, DSA, immunosuppression with calcineurin inhibitor (CNI) versus mammalian target of rapamycin (mTOR) inhibitor, steroid free versus steroid-based immunosuppression, age, and time since transplant. For history of rejection or DSA, a patient was classified on the first event date and moving forward. For CNI versus mTOR, and steroid free versus steroid-based immunosuppression, the variables were coded daily based on the prescription records in the EHR.

Analysis

To assess clinical outcomes in association with MAPS implementation, we analyzed system-level performance using SPC and patient-level rejection risk using a multivariable parametric survival model to control for changes in covariates over time.

System-Level Analysis

Because MAPS is a practice change that affects the care and related health outcomes of all patients within the system,

Table 1. Medication Adherence Promotion System Core Components

System Component (Key Driver)	Description
Adherence promotion training for clinical team members training	Clinical psychologists specializing in adherence behavior change train clinical team members to use nonjudgmental communication and problem solving to coach patients in identifying and implementing strategies to overcome adherence barriers. Training is conducted in two 60-minute sessions and includes didactics via slide presentation, role-plays, and scripts to guide discussions.
Adherence risk screening during multidisciplinary previsit planning[21]	Patients are screened for adherence risk during weekly previsit planning meetings using data extracted from the EHR. Patients are flagged as at risk for nonadherence if they: • Have high variation in tacrolimus drug levels (>2 SD across 4 levels). • Lack clinical follow-up within recommended time frame (<4 mo). • Lack laboratory monitoring within recommended time frame (<6 wk).
Assessment of barriers to medication adherence[21]	During clinical intake, patients and/or caregivers complete the barriers assessment tool, a 14-item checklist of common adherence barriers that asks, "What gets in the way of you taking your immunosuppression medication?" (see Fig 2). The assessment is administered semiannually via tablet computer to caregivers of patients < 10 years old and to patients and caregivers for patients ≥ 10 years old. Responses are transmitted to the EHR and prompt an alert for clinicians to review before entering the clinic room.
Designation of team member responsibility	Team members are designated to have specific responsibility for partnering with the patient and/or caregiver to address each potential barrier (see Fig 2).
Action planning with patients (see Table 2)	During the clinic visit, the designated team member discusses any identified adherence barriers with the patient and/or family, and together they develop an action plan to overcome identified barriers. • For the most common barriers (eg, forgetting, dislike the taste, pill swallowing difficulties, side effects, embarrassment of taking medicine), evidence-based shared decision-making tools were created using input from families, the hospital family advisory board, and patient education committee (see Table 2). Transplant coordinators review these tools with the patient and/or caregiver and a problem-solving approach is used to identify one solution for the patient to try with a clear plan for the who, what, how, where, and when this solution could be implemented. • For other treatment-related barriers (eg, too many medications, don't think they work), the nephrologist, nurse practitioner, or clinical pharmacist provides education and reviews alternative options, including simplifying the treatment regimen to once-daily preparations of tacrolimus and azathioprine. • For resource-related barriers (eg, ran out, can't afford) and health literacy, the social worker is consulted. • For psychological adherence barriers (eg, oppositional behaviors, anxiety, depression, executive functioning deficits), the patient is referred to psychology.
Follow-up assessment and adaptation	The team member who is partnered with the patient and/or caregiver to develop an adherence action plan is responsible to follow up within 2 weeks to assess progress with the action plan and adapt as appropriate.
Optional electronic adherence monitoring	Patients deemed high risk by the clinical team based on EHR-based adherence risk screen, clinical judgment, and rejection events are offered electronic adherence monitors (ie, SimpleMed Pillboxes, and AdhereTech bottles) for real-time tracking of adherence data. Data are reviewed with patients during follow-up clinic visits and integrated with barriers assessment and action planning as described. Additionally, if real-time data suggest lack of device use for 3 days, transplant coordinators reach out to families to see if the device is working properly, if the patient is using the device, and/or if they are having difficulties taking their medicine. A nonjudgmental approach is used, with coordinators calling out of concern to partner with patients to overcome any identified barriers to medication taking or device use.

Abbreviation: EHR, electronic health record.

we chose methods used in health care to track system performance in avoiding adverse health outcomes over time, such as malignancy rates in KT recipients[23] or osteomyelitis treatment failures in children.[24] We performed an interrupted time series analysis using SPC to assess the frequency of late rejection over time in the popultion.[25,26] We used both an "events between" chart (G-chart), especially suited to detect decreases in rare events, and a U-chart to assess average monthly rejection rates before and after MAPS.

For the G-chart analysis we evaluated patient-days between late acute rejection episodes in the population according to the published methods.[27,28] Each rejection episode was plotted as a data point on the *x*-axis by date and on the *y*-axis by the number of patient-days in the population since the prior rejection occurred. A median and 3 SD upper control limit (UCL) were calculated based on the geometric distribution of the data using the first 25 data points of the pre-MAPS period and carried forward for comparison of future system performance.[27] Variation of

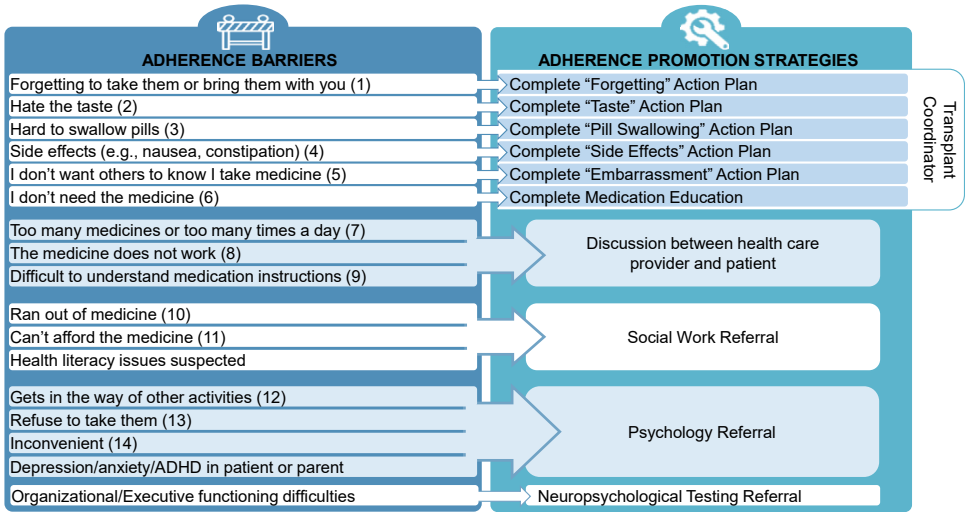


Figure 2. Algorithm to assign responsibility for partnering with patients to resolve specific adherence barriers. Each adherence barrier that a patient might identify was assigned to a specific team member who is responsible to partner with the patient to develop an action plan to overcome the identified barrier. In this manner there was clarity of responsibility in the clinic as to who should work with a patient to address each barrier. These barriers were assessed for caregivers if the patient was < 10 years of age, and for both patients *and* caregivers (if present) if the patients was ≥ 10 years of age.

data within projected control limits was considered "common cause" variation, related to random effects, whereas data points outside the control limits or displaying other statistically unlikely patterns (eg, ≥8 consecutive data points above or below the centerline) were considered "special cause," indicating evidence of change in system performance.[24,27] The industry standard to provide a reliable range for expected future data given the past data is 3 SD UCL.[26,27] When a special cause was noted, the centerline and UCL were recalculated as of that time point to estimate performance of the new system.[27,28]

A U-chart was used to evaluate change in monthly rejection rates before and after implementation of MAPS. Data points for this chart were plotted by month on the *x*-axis and on the *y*-axis as the number of observed rejections in the given month divided by the number of observed 100 patient-months in that month. Calculations for mean and 3 SD UCL and lower control limit (LCL) for this chart were based on the Poisson distribution of rate data.[26] Standard special cause criteria of 8 consecutive data points below the centerline or any data point outside the control limits were considered evidence of improvement in outcome.[26]

Patient-Level Analysis

Although SPC is an accepted method of assessing change in system performance over time,[23-28] it lacks the ability to control for changes in case-mix risk factors (covariates) that might also influence observed changes in outcome. We therefore conducted analyses to examine whether the incidence rate of rejection differed across the MAPS implementation period after adjusting for covariates. Because the observation time was not uniform and some covariates varied over time for a given patient, we used a multivariable parametric survival model[29-31] rather than a Poisson model. The hazard ratio in this model can be interpreted as an adjusted incidence rate ratio (IRR). To account for dependence that resulted from some patients having repeated rejections (within and/or across both MAPS periods), we used a model with shared frailty.[29-31] A shared frailty survival model is analogous to a random effects regression model with a patient-specific random intercept. We used an exponential survival distribution because it demonstrated a better fit to the data relative to other probability distributions (eg, Weibull).

To develop a final model with at least 5 rejection events per covariate,[32] we first performed univariable survival analysis with each of 12 potential covariates (Table 3). Covariates with fewer than 5% missing values and either an incidence rate ratio of >1.3 (or <0.77) or $P < 0.1$ were included in the final multivariable model. We also used shared frailty, exponential survival models to evaluate whether the incidence of biopsies, rejections per biopsy, de novo DSA formation, and type of rejection differed between the pre- and post-MAPS periods. The *P* values for these analyses indicate whether the incidence rate for a variable (eg, biopsies) differed across MAPS implementation periods. Stata (StataCorp) was used for all analyses, and $P < 0.05$ using a 2-tailed test was considered statistically significant.

Results

Characteristics of Cohort

Over 4 years, we assessed 121,133 active patient-days in 130 patients. Seventy-two patients met the inclusion

Table 2. Medication Taking Shared Decision-Making Action Plans

Shared Decision- Making Action Plan	Description
Forgetting action plan	Provides both technology-based strategies (such as setting phone reminders or using an app reminder) and non-technology-based strategies (pairing medication-taking with daily routines such as brushing teeth or eating) to address forgetting to take medication.
Pill swallowing action plan	Provides various strategies to make pill swallowing easier, including using the candy practice method, changing head positions, and coating the medicine.
Taste action plan	Provides different methods that could be used to mask the taste, such as the pharmacist flavoring liquid medicines, using coatings or syrups, numbing taste buds before taking with popsicle/ice, or mixing medicine with foods (if possible). On the back of this handout, there are guidelines developed by the pharmacist on whether medications could be cut or crushed, or need to be swallowed whole.
Embarrassment action plan	Suggests ways to store and take medications discreetly (eg, small bag in purse or take in the bathroom or in car), as well as ways to talk about medicine with peers (eg, develop a script about the kidney disease and need for medicine).
Side effects action plan	Created primarily for providers to quickly examine the most common side effects for most transplant-related medications and to see if medications can be changed, medications can be added to address side effects, or doses can be changed to reduce side effects.

criteria at the beginning of the observation period. Fifty-eight patients entered the cohort during the study (49 new transplants and 9 transfers), and 43 left the cohort due to transferred care (33), allograft failure (4), loss to follow up (3), death (2), or malignancy (1). All allograft failures were in the pre-MAPS period. We observed 51 total rejections (34 before MAPS vs 17 after MAPS). Twenty-four patients had 1 rejection, 4 patients had 2, 5 patients had 3, 1 patient had 4. Of all patient-days of observation among those with multiple rejections, the proportion occurring in the pre- and post-MAPS periods were similar (51.9% vs 48.1%), which was comparable to the entire sample (51.6% of total patient-days of observation occurred before MAPS vs 48.4% after MAPS). Table 3 depicts the patient and transplant characteristics for the sample and the time at risk for each characteristic according to MAPS period.

Pre-MAPS Implementation

During the 26-month baseline and implementation period, there was a median of 1,700 (UCL, 7,939) patient-days between rejection episodes in the population. The average monthly late acute rejection rate before MAPS implementation was 1.61 rejections per 100 patient-months. There were no special causes (see Methods), suggesting a stable system with predictable performance (Fig 3).

Post-MAPS Implementation

In the 5 weeks after MAPS implementation, there were 4 rejection episodes (Fig 3). After April 7, 2016, we observed 11,658 patient-days between rejection episodes above the previous UCL of 7,939. There was an additional special cause of 9,464 patient-days noted on August 31, 2017. Adjustment of median and UCL after the special cause demonstrated an increased median of 3,240 (UCL, 16,715) patient-days between rejection episodes. After MAPS implementation, the average monthly late acute rejection rate fell from 1.61 to 0.88 rejections per 100 patient-months. Notably, all but 3 of the 22 monthly data points after MAPS implementation fell below the previous monthly average rejection rate, including the final 12 consecutive months indicating special cause on the U-chart. There were an estimated 14 fewer rejections in the 22-month postimplementation period than predicted by the prior 2 years.

Barriers Assessment and Interventions

After MAPS implementation, 418 barriers assessment tools were completed for 99 out of 110 patients (90%) in the post-MAPS period. Barriers were identified for 36 of 99 patients (36%) on 116 of 418 assessments (27.8%). The most common barriers were side effects (19.8%), disliking the taste (19.8%), and forgetting (16.4%). A total of 91 interventions were documented in 29 patients. Twenty-eight patients were offered adherence monitoring, but 6 (21%) refused, and 6 (21%) had technical difficulties. Sixteen participants (57%) successfully used adherence monitors during the study period (median of 403.5 [range, 19-804] days).

Univariable and Multivariable Analysis

In univariable survival analysis, MAPS was the only variable associated with decreased rejection rate, with an IRR of 0.51 (95% CI, 0.28-0.94; $P = 0.03$) (Table 3). DSA history, repeat KT, donor type, race, prior rejection, HLA mismatch, and time since transplant all met the criteria for inclusion in the final model. None of the covariates significantly violated the proportionality assumption. As shown in Table 4, after adjusting for covariates in the multivariable model, the post-MAPS treatment period was associated with a 50% reduction in the incidence of rejection (IRR, 0.50 [95% CI, 0.27-0.91]; $P = 0.02$). DSA history was associated with a >2-fold increase in incidence of rejection (IRR, 2.27 [95% CI, 1.12-4.57]; $P = 0.02$), and each year after transplant was associated with a 13% decreased incidence of rejection (IRR, 0.87 [95% CI, 0.77-0.97]; $P = 0.02$).

Table 3. Sample Characteristics and Univariable Analysis

Characteristics	No. of Patients	No. of Patient-Days		Univariable Survival Analysis IRR (95% CI)	*P*
		Pre-MAPS	Post-MAPS		
Total no. of patients and patient-days	130 (100%)	62,610 (100%)	58,523 (100%)	0.51 (0.28-0.93)[a]	0.03
Sex					
Male	73 (56.2%)	36,164 (57.8%)	30,765 (52.6%)		
Female	57 (43.9%)	26,446 (42.2%)	27,758 (47.4%)	1.18 (0.57-2.42)	0.7
Race/ethnicity					
Non-Hispanic White	93 (71.5%)	46,056 (73.6%)	40,190 (68.7%)		
Minority	37 (28.5%)	16,554 (26.4%)	18,333 (31.3%)	0.70 (0.30-1.66)	0.4
Black	26 (20.0%)				
Asian	3 (2.3%)				
Other	1 (0.8%)				
Hispanic	7 (5.4%)				
Donor type					
Living	86 (66.2%)	45,146 (72.1%)	39,348 (67.2%)		
Deceased	44 (33.9%)	17,464 (27.9%)	19,175 (32.8%)	0.71 (0.31-1.63)	0.4
Repeat transplant					
No	121 (93.1%)	59,072 (94.4%)	55,935 (95.6%)		
Yes	9 (6.9%)	3,538 (5.7%)	2,588 (4.4%)	2.33 (0.69-7.87)	0.2
HLA mismatch					
0/6-3/6	82 (63.1%)	41,799 (66.8%)	35,404 (60.5%)		
4/6-6/6	46 (35.4%)	19,231 (30.7%)	22,262 (38.0%)	1.34 (0.64-2.82)	0.4
Missing	2 (1.5%)	1,580 (2.5%)	857 (1.5%)		
Induction					
Basiliximab	102 (78.5%)	49,442 (79.0%)	47,640 (81.4%)		
Thymoglobulin/OKT3/ none	16 (12.3%)	6,254 (10.0%)	4,947 (8.5%)	1.41 (0.45-4.35)	0.6
Missing	12 (9.2%)	6,914 (11.0%)	5,936 (10.1%)		
CNI					
No	14 (10.8%)	16,092 (25.7%)	5,810 (9.9%)		
Yes	116 (89.2%)	46,518 (74.3%)	52,713 (90.1%)	1.10 (0.44-2.72)	0.8
Steroids					
No	53 (40.8%)	40,480 (64.7%)	34,791 (59.5%)		
Yes	77 (59.2%)	22,130 (35.4%)	23,732 (40.6%)	1.10 (0.53-2.25)	0.8
DSA history					
No	86 (66.2%)	45,594 (72.8%)	38,715 (66.2%)		
Yes	44 (33.9%)	17,016 (27.2%)	19,808 (33.9%)	1.77 (0.88-3.56)	0.1
Prior rejection					
No	106 (81.5%)	46,974 (75.0%)	49,387 (84.4%)		
Yes	24 (18.5%)	15,636 (25.0%)	9,136 (15.6%)	2.05 (0.92-4.58)	0.08
Age, y[b]	15.5 [7-20]	16 [7-20]	15 [8-20]	1.03 (0.98-1.09)	0.2
Years from transplant[b]	1.95 [1.0-5.2]	2.62 [1.0-6.7]	4.08 [2.2-7.1]	0.90 (0.80-1.01)	0.07
Years to rejection	2.93 [2.0-5.2]	2.49 [1.9-4.1]	3.38 [2.4-5.8]		

Values given as count (percentage), median [interquartile range], or IRR (95% CI).
Abbreviations: CNI, calcineurin inhibitor; DSA, donor-specific antibody; HLA, human leucocyte antigen; IIR, incidence rate ratio; MAPS, Medication Adherence Promotion System; OKT3, mitogen anti-CD3 monoclonal antibody; PRA, panel-reactive antibody.
[a]IRR and 95% CI for late rejection in post-MAPS period versus pre-MAPS.
[b]As of the start of analysis.

Secondary Analysis and Balancing Measures

As Table 5 shows, rate of kidney biopsies per 100 patient-months did not differ significantly before and after MAPS implementation (3.17 vs 2.91; $P = 0.5$) but the rate of biopsies positive for rejection decreased significantly from 50% to 30% ($P = 0.03$). Thirteen patients (0.6 events per 100 patient-months) developed de novo DSA during the pre-MAPS period compared with 7 patients (0.4 events per 100 patient-months) in the post-MAPs period ($P = 0.3$). The severity and type of rejection did not differ significantly before and after MAPS implementation ($P = 0.8$).

Discussion

Although several clinical trials have demonstrated the efficacy of multicomponent interventions to improve

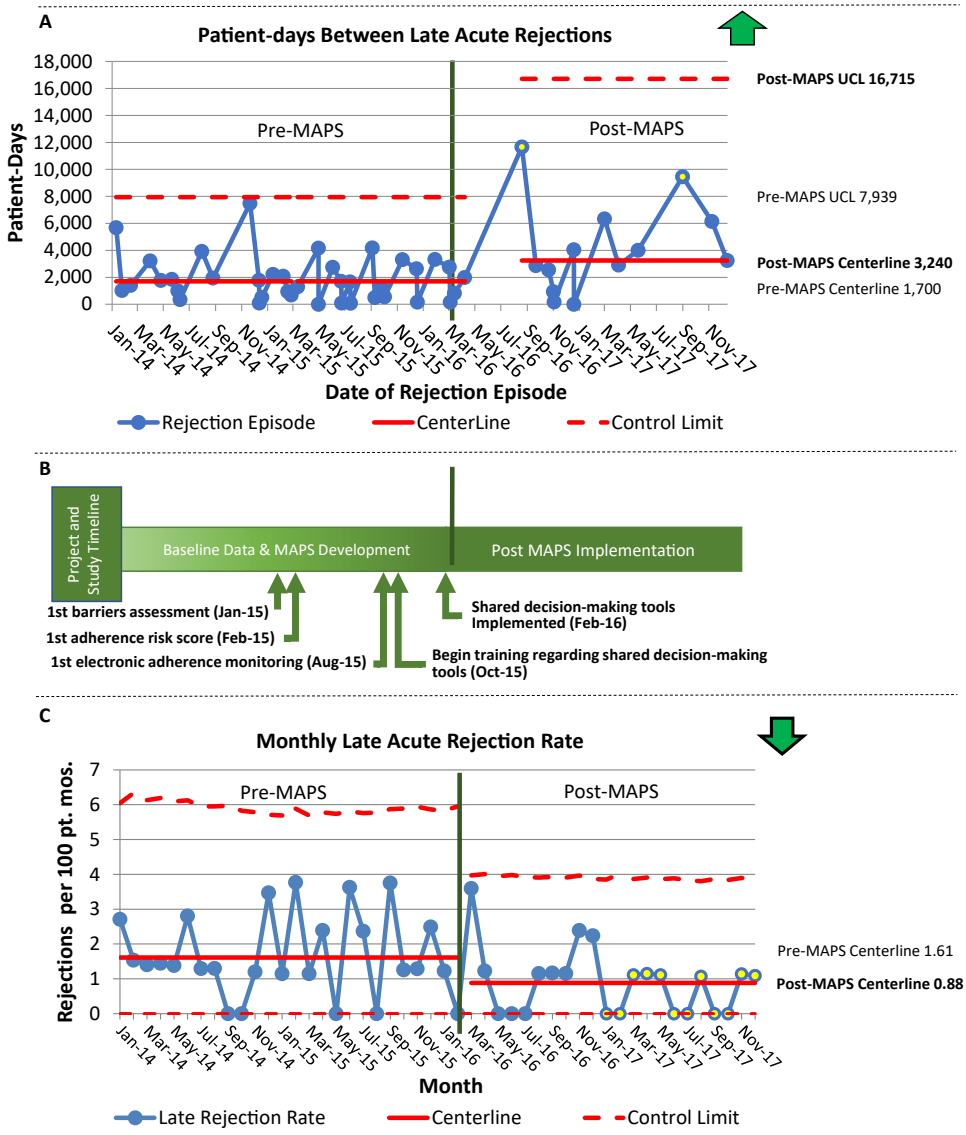


Figure 3. Statistical process control analysis of late acute rejection before and after MAPS. The phases of project implementation are depicted in the graphic (B) for comparison with the statistical process control time series analysis (A, C). *B* indicates the timeline for the project, including the baseline data collection and MAPS planning. The solid vertical line represents the date at which full MAPS implementation occurred (March 2016). (A) The G-chart and (C) U-chart are statistical process control charts of late acute rejection aligned with the timeline in panel *B*. The solid red line indicates the center line, and the dotted red lines indicate the UCL and LCL. The green arrow indicates desired direction of change. The data points in yellow represent "special cause" or evidence of a changed (improved) system. *A* depicts a G-chart for patient-days between rejection episodes based on the geometric distribution. Each rejection episode is plotted on the chart with the date on the *x*-axis and the patient-days since the prior rejection on the *y*-axis. There were no special causes during the baseline and development phase, indicating a stable system. The first special cause was noted starting 5 weeks after implementation and continuing for the next 11,658 patient-days. There was a second special cause of 9,464 patient days several months later. Adjustment of the median and UCL for data after the special cause indicated an increase in median patient days between rejections from 1,700 to 3,240. *C* depicts a U-chart for the rejections per 100 patient-months. There is a data point for each month of observation along the *x*-axis and the number of rejections in that month divided by 1/100th of the number of patients observed during that month (150 patients followed for 1 months would be a denominator of 1.5). Centerlines, UCLs, and LCLs where calculated for comparison between the pre-MAPS phase and the post MAPS phase. Pre-MAPS average monthly rejection rate per 100 patient-months was 1.61 compared to 0.88 post-MAPS. Of note, all but 2 of the post-MAPS data points are below the prior centerline, including the last 12 consecutive data points indicating special cause and evidence of improvement. Abbreviations: LCL, lower control limit; MAPS, Medical Adherence Promotion System; pt. mos., patient-months; UCL, upper control limit;

Table 4. Multivariable Survival Analysis: Incidence Rate Ratio of Late Acute Rejection

Risk Factor	IRR (95% CI)	*P*
Post-MAPS vs pre-MAPS	0.50 (0.27-0.91)	0.02
DSA history	2.27 (1.12-4.57)	0.02
Repeat kidney transplant	1.65 (0.53-5.16)	0.4
Deceased donor	0.58 (0.23-1.45)	0.2
Minority status	0.56 (0.24-1.34)	0.2
Prior rejection	1.37 (0.65-2.91)	0.4
HLA mismatch	1.41 (0.63-3.20)	0.4
Time since transplant, per year	0.87 (0.77-0.97)	0.02

Abbreviations: CNI, calcineurin inhibitor; DSA, donor-specific antibody; HLA, human leucocyte antigen; IIR, incidence rate ratio; MAPS, Medication Adherence Promotion System.

adherence in highly structured research environments,[11,14,15] to our knowledge ours is the first to implement such interventions in clinical practice and observe an associated decrease in late acute rejection. The novelty of our approach was to adapt techniques typically limited to research contexts for use in clinical practice and reliably implement them using EHR automation and existing clinical staff.

MAPS was fully integrated as standard care in our clinic and was applied to all patients on immunosuppression, thus demonstrating its feasibility in a real-world setting. Using a system/population-based analysis with statistical process control, we observed a 45% reduction in late acute rejection rate within months of complete MAPS implementation. This reduction was sustained for nearly 2 years, despite no significant change in kidney biopsy rates.

Univariable and multivariable analysis evaluating patient-level outcomes and adjusting for important covariates found that MAPS implementation was associated with a 50% lower incidence of rejection, suggesting that the observed decrease in late acute rejection is not explained by known changes in case-mix. In this analysis, both time since transplant and presence of DSA were significant predictors of rejection risk, consistent with the literature.[33] In the 22 months after MAPS implementation, 14 fewer late rejections were observed than expected based on results from the prior 2 years.

We primarily observed a decrease in TCMR, which is more amenable to treatment and is thought to have less negative impact on allograft survival than AMR. However, improvement in late acute rejection has been shown to decrease chronic rejection and graft failure in both children[34,35] and adults,[36,37] and recent literature has identified deleterious effects of TCMR on long-term graft function.[3] Furthermore, treatment of rejection often requires hospitalization and may include invasive procedures.[38] Thus, the observed decrease in late acute rejection is likely to have an important impact on allograft survival, health care cost, and quality of life. Although there was a decrease in de novo DSA after MAPS implementation, it was not statistically significant, and no difference in AMR was noted.

This work builds on the landmark randomized clinical trials[11,14,15] that demonstrated the efficacy of multicomponent interventions[16] to improve medication adherence in transplantation. These interventions included feedback of adherence data, social support, problem solving to overcome adherence barriers with staff members trained in developing action plans with patients, and follow-up evaluations to iteratively modify and improve adherence. Although the "dose" of these adherence interventions varied by study, all showed adherence improvements. Notably, each study highlighted the future need to implement such approaches in the clinical environment and to understand impact on rejection rates.

Thus, our QI study was not designed to directly measure adherence nor to test the efficacy of such interventions, but rather to implement them in clinical care and measure the associated changes in rejection rates. Our approach builds on well-established models of behavior change in children, adolescents, and young adults using the key components of problem solving with trained team members, action planning to address adherence barriers, follow-up evaluations to modify action plans as needed, and engagement of social support. Due to cost and

Table 5. Biopsy Results, Rejections per Biopsy, and De Novo DSA

	Total	Pre-MAPS	Post-MAPS	*P*
No. of patient-months	3,982	2,058	1,924	
No. of biopsies (rate per 100 patient-months)	122 (3.05)	66 (3.17)	56 (2.91)	0.5
No. of late acute rejection (per 100 patient-months)	50 (1.25)	33 (1.58)	17 (0.88)	0.03
Ratio of no. of late acute rejections to biopsies	0.41	0.50	0.30	0.03
No. developing de novo DSA (rate per 100 patient-months)	20 (0.5)	13 (0.6)	7 (0.4)	0.3
Rejection type/grade				0.8
Acute TCMR Banff 1a/b	43 (86%)	29 (88%)	14 (82%)	
Acute TCMR Banff 2a	1 (2%)	1 (3%)	0 (0%)	
Mixed Rejection	3 (6%)	1 (3%)	2 (12%)	
Active ABMR	3 (6%)	2 (6%)	1 (6%)	

P values indicate whether the incidence rate for a variable (eg, biopsies) differ across pre-MAP and MAPS.
Abbreviations: ABMR, antibody-mediated rejection; DSA, donor-specific antibody; MAPS, Medication Adherence Promotion System; TCMR, T-cell-mediated rejection.

logistical challenges, most patients in our study did not use electronic monitoring. Indeed, fewer than two-thirds of the 28 patients who were offered electronic monitoring found it acceptable and functional in practice. This suggests that further innovation is necessary before this becomes a widespread intervention in clinical practice.

It is informative to note the similarity in outcomes between MAPS and TAKE-IT,[11] a randomized clinical trial of adherence interventions in adolescent KT recipients. The TAKE-IT study was underpowered to assess rejection outcomes, but the difference in rejection rates between the control and treatment groups in TAKE-IT was remarkably similar to our pre- and post-MAPS implementation rejection rates. Whereas TAKE-IT observed 1.69 rejections per 100 patient-months in the control arm compared with 1.06 in the treatment arm ($P = 0.30$),[11] we observed a reduction from 1.61 rejections per 100 patient-months at baseline to 0.88 in the 22-month period after MAPS implementation—a result that was statistically significant in multivariable survival analysis after controlling for other rejection risk factors.

Several factors are likely required for MAPS to be successful, and it is difficult to identify the key ingredient. However, the use of evidence-based adherence strategies that foster a culture of nonjudgmental problem solving, involvement of interdisciplinary team members with complementary expertise, creation of algorithms to systematically assess and address adherence barriers by the clinical team, and EHR to facilitate processes reliably all likely contributed. To adapt this approach to other settings, providers must assign responsibility for addressing adherence barriers among existing staff and empower them with training and tools to address adherence barriers. Work is underway to adapt and spread MAPS in KT programs throughout the United States via the Improving Renal Outcomes Collaborative (IROC) as well as through learning health care networks for other chronic conditions (eg, epilepsy and rheumatology).

We note several limitations that impact the generalizability of this study. First, adherence was not directly measured in most patients due to a lack of resources and patient acceptance. However, the assessment of an important clinical end point (eg, rejection) is a strength that offsets this limitation. Second, the findings are from a single center, and generalizing MAPS to other environments will require adaptation to fit the local context. Third, inference of causality in observational studies should be undertaken with caution. Although every attempt was made to control for known covariates, in time-series analysis it is not possible to control for all possible confounders, including secular trending in outcomes, that may influence results. Nonetheless, interrupted time series is an appropriate methodology to evaluate the clinical impact of interventions already shown to be effective through clinical trials,[25] and the observation of a large treatment effect that corresponds with the timing of the interventions, persists for 2 years, and corroborates prior research findings (eg, TAKE-IT) all increase confidence in the conclusions. Finally, due to the relatively small sample size and rarity of rejection, we are unable to assess which adherence barriers or intervention were most associated with the risk of rejection.

In summary, we adapted evidence-based adherence interventions and reliably implemented them at the point of care with clinical staff supported by the EHR. This multimodal, provider-delivered system to identify and mitigate patient and caregiver barriers to taking immunosuppression was associated with a 50% reduction in late allograft rejection. MAPS has become the standard of care in our clinic and should be tested in other settings.

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